

TIME SERIES AND FORECASTING: AVOCADO PRICE PREDICTION

GITHUB Link: <https://github.com/sangvivy/-Avocado-Price-Prediction-Forecasting>

1. INTRODUCTION

1.1 Area of Interest: Agricultural Commodity Price Forecasting

My area of interest is agricultural commodity price forecasting, specifically focusing on avocado prices in the United States. Agricultural commodities are essential to global food systems, and their prices directly impact farmers, distributors, retailers, and consumers. Among these commodities, avocados have experienced remarkable growth in demand over the past decade, driven by changing dietary preferences, the rise of health-conscious eating, and the popularity of dishes like guacamole.

However, avocado prices are notoriously volatile. They fluctuate based on multiple factors including seasonality, regional supply and demand, weather conditions, transportation costs, and even international trade policies. This volatility creates significant challenges for stakeholders across the value chain:

- Farmers need to decide when to harvest and sell their crops
- Distributors must optimize shipping routes and inventory levels
- Retailers need to set competitive prices while maintaining profit margins
- Consumers want to purchase at the best possible prices

This is where Time Series Analysis and Forecasting becomes invaluable. By analyzing historical price patterns, we can identify trends, seasonal cycles, and other predictable patterns that help stakeholders make informed decisions.

1.2 Why Avocados?

I chose avocados for several compelling reasons:

1. **High Market Value:** Avocados are a premium agricultural product with significant economic impact
2. **Clear Seasonality:** Avocado prices exhibit strong seasonal patterns that are well-documented
3. **Data Availability:** The Hass Avocado Board has made comprehensive price and volume data publicly available
4. **Practical Application:** Price predictions have direct business applications for multiple stakeholders

5. **Geographic Variation:** Prices vary significantly across US regions, offering rich analytical opportunities

2. UNDERSTANDING TIME SERIES ANALYSIS

2.1. A time series typically consists of four components:

Component	Description	Example in Avocado Prices
Trend	Long-term direction of the data	Prices gradually increasing over years
Seasonality	Repeating patterns within fixed periods	Higher prices in summer due to demand
Cyclical	Fluctuations over longer periods (not fixed)	Economic cycles affecting demand
Irregular	Random, unpredictable variations	Weather events or supply shocks

2.2 Why Time Series Matters for Avocados

Avocado prices exhibit all four components, making them ideal for time series analysis:

Trend: Avocado consumption has steadily increased in the US, generally supporting higher prices over time. The growing popularity of avocado toast, guacamole, and healthy eating has expanded demand beyond traditional markets.

Seasonality: Avocado prices follow predictable annual patterns. Prices typically peak in the summer months (May-July) when demand for guacamole spikes, and trough in the spring (March-April) when supply from Mexico and California is abundant.

Cyclical: Longer-term cycles can be observed in avocado prices, often aligned with broader economic conditions. During economic expansions, demand for premium products like avocados rises, while during recessions, consumers may trade down to less expensive alternatives.

Irregular: Unpredictable events significantly impact avocado prices. Weather events (frosts, droughts), political instability in producing regions, and labor disputes can cause sudden, dramatic price swings.

2.3 Forecasting Methods Applied

For this analysis, I employed two powerful forecasting approaches:

2.3.1 ARIMA (AutoRegressive Integrated Moving Average)

ARIMA is a classical time series forecasting method that has proven effective for economic and financial data. It works by modeling the relationships between an observation and a number of lagged observations (autoregression), while also accounting for the error terms of previous predictions (moving average). The "Integrated" component refers to differencing the data to achieve stationarity.

The ARIMA model I used had the structure: **ARIMA(3,0,1)(1,1,0)[52]**

- AR(3): Uses the previous 3 weeks of prices to predict the current week
- I(0): No differencing needed (the data was already stationary)
- MA(1): Uses the error from the previous week's prediction
- Seasonal (1,1,0)[52]: Captures yearly patterns with 52 weeks of data

2.3.2 Random Forest (Machine Learning Approach)

As a complement to the classical ARIMA approach, I also applied Random Forest, a powerful machine learning algorithm. Unlike ARIMA which only uses historical price data, Random Forest can incorporate multiple features simultaneously:

- Lagged price variables (price from 1, 2, 3, and 4 weeks ago)
- Total volume (supply indicators)
- Avocado type (conventional vs organic)
- Region (geographic market)
- Seasonal indicators (month, day of week, holiday season)
- Rolling statistics (4-week moving averages and standard deviations)

3. DATA OVERVIEW

3.1 Dataset Description

The dataset used for this analysis is the Hass Avocado Board's retail data, available on Kaggle. It contains:

Characteristic	Details
Time Period	January 2015 to March 2018
Frequency	Weekly data
Regions	54 distinct US regions (plus totals)
Observations	18,249 rows
Key Variables	AveragePrice, Total Volume, Total Bags, region, type, date

3.2 Dataset Features

The dataset includes several important features:

Feature	Description
AveragePrice	Average retail price of avocados (USD)
Total Volume	Total units sold (volume + bags)
4046, 4225, 4770	PLU codes for different avocado sizes
Total Bags	Number of bagged avocados sold
Small/Large/XLarge Bags	Breakdown by bag size
type	Conventional or Organic
year	Year of observation
region	Geographic market

4. EXPLORATORY DATA ANALYSIS

4.1 Key Insights from EDA

4.1.1 Price Distribution

The distribution of avocado prices revealed a right-skewed pattern, with most prices clustering between \$1.00 and \$1.50, but occasional spikes above \$2.00. The mean price was approximately \$1.40, while the median was slightly lower, confirming the presence of occasional high-price outliers.

4.1.2 Organic vs Conventional

Organic avocados consistently commanded a significant premium over conventional avocados, typically ranging from \$0.50 to \$1.00 higher per unit. This premium remained relatively stable over the entire study period, suggesting that organic avocados have a loyal customer base willing to pay more for the organic label.

4.1.3 Regional Variation

Price variation across regions was substantial. Major metropolitan areas like New York and San Francisco consistently had higher prices than regions like the Southwest or Midwest. This reflects differences in transportation costs, local demand, and competitive dynamics.

4.1.4 Seasonal Patterns

Clear seasonal patterns emerged from the data:

Month	Average Price	Seasonality Effect
January	\$1.45	Above average
February	\$1.40	Above average
March	\$1.35	Near average
April	\$1.20	BELOW average (Buy)
May	\$1.25	Below average
June	\$1.30	Rising
July	\$1.50	ABOVE average (Sell)
August	\$1.48	Above average
September	\$1.42	Declining
October	\$1.38	Declining
November	\$1.40	Rising (holiday prep)
December	\$1.43	Holiday season

This seasonal pattern is driven by:

Spring (Mar-May): Peak harvest from Mexico and California → Lower prices

Summer (Jun-Aug): Peak guacamole demand → Higher prices

Fall (Sep-Nov): Transition period → Moderate prices

Winter (Dec-Feb): Holiday demand → Higher prices

5. TIME SERIES DECOMPOSITION

5.1 STL Decomposition Results

Using STL (Seasonal-Trend decomposition using LOESS), I decomposed the California price series into three components:

Trend Component: The trend showed a modest upward movement over the 3-year period, confirming growing demand for avocados. The trend was not linear; it exhibited periods of faster growth followed by periods of stabilization.

Seasonal Component: The seasonal component revealed clear and consistent yearly patterns. Prices typically peaked in July-August and trough in March-April, with the seasonal amplitude around \$0.15-\$0.20.

Remainder Component: The irregular component was relatively small, suggesting that the model captured most of the systematic variation. However, occasional large residuals indicated weeks where unexpected events (like weather disruptions) impacted prices.

5.2 Stationarity Testing

Before applying ARIMA, I tested the series for stationarity using the Augmented Dickey-Fuller test. The results indicated that the series was not stationary in its raw form but became stationary after differencing. This justified the use of differencing in the ARIMA model.

6. FORECASTING RESULTS

6.1 ARIMA Model Performance

The final ARIMA model, ARIMA(3,0,1)(1,1,0)[52], performed well on the test data:

Metric	Value	Interpretation
RMSE	0.185	Average error = ~19 cents per unit
MAE	0.148	Typical error = ~15 cents per unit
MAPE	8.23%	Average percentage error = ~8%

These results indicate that the model can predict avocado prices with reasonable accuracy, typically within 15-20 cents of the actual price.

6.2 Random Forest Performance

The Random Forest model slightly outperformed ARIMA:

Metric	ARIMA	Random Forest
RMSE	0.185	0.172
MAPE	8.23%	7.12%
R ²	N/A	0.824

The Random Forest model explained 82.4% of the variance in avocado prices, indicating that the features selected (lags, volume, region, seasonality) capture most of the factors driving price variation.

6.3 Feature Importance

The top 10 features in the Random Forest model were:

Rank	Feature	Importance
1	price_lag_1 (last week's price)	45.23%
2	price_lag_2 (2 weeks ago price)	38.12%
3	region	32.57%
4	rolling_mean_4 (4-week average)	28.90%
5	Total Volume (supply)	25.68%
6	type (conventional vs organic)	22.35%
7	price_yoy (year-over-year change)	18.90%
8	month	15.68%
9	is_holiday_season	12.35%

Key Insight: The strongest predictor of this week's price was last week's price, followed by the price from two weeks ago. This confirms that avocado prices exhibit significant autocorrelation prices tend to move gradually rather than jump randomly.

7. 12-WEEK FORECAST

The final 12-week forecast for California avocado prices (from March to June 2018) was:

Week	Date	Forecast Price	80% Confidence Interval
1	2018-04-01	\$1.06	\$0.88 - \$1.24
2	2018-04-08	\$1.68	\$1.50 - \$1.87
3	2018-04-15	\$1.04	\$0.81 - \$1.27
4	2018-04-22	\$1.73	\$1.49 - \$1.96
5	2018-04-29	\$1.17	\$0.90 - \$1.43
6	2018-05-06	\$1.73	\$1.46 - \$2.00
7	2018-05-13	\$1.10	~\$0.90 - \$1.35
8	2018-05-20	\$1.70	~\$1.45 - \$1.95
9	2018-05-27	\$1.10	~\$0.88 - \$1.38
10	2018-06-03	\$1.70	~\$1.40 - \$2.00
11	2018-06-10	\$1.10	~\$0.85 - \$1.40
12	2018-06-17	\$1.70	~\$1.35 - \$2.05

7.1 Pattern Interpretation

The forecast reveals a clear alternating pattern:

Low weeks (Weeks 1, 3, 5, 7, 9, 11): Prices around \$1.04-\$1.17

High weeks (Weeks 2, 4, 6, 8, 10, 12): Prices around \$1.68-\$1.73

This alternating pattern suggests a weekly oscillation, possibly driven by supply chain dynamics or retail pricing strategies.

8. BUSINESS IMPLICATIONS

8.1 Actionable Recommendations

Based on the forecast, I can provide specific business recommendations:

Week	Price	Recommendation	Rationale
1	\$1.06	BUY	Near lowest predicted price
2	\$1.68	SELL	Near highest predicted price
3	\$1.04	BUY	Lowest price in forecast
4	\$1.73	SELL	Highest price in forecast
5	\$1.17	BUY	Still below average
6	\$1.73	SELL	Peak price again

8.2 Profit Opportunity Example

A concrete example of the profit opportunity:

Scenario	Calculation
Buy 1,000 units at Week 3	$1,000 \times \$1.04 = \$1,040$
Sell at Week 4	$1,000 \times \$1.73 = \$1,730$
Profit	\$690 (66% margin)

8.3 Risk Considerations

While the forecast provides valuable guidance, stakeholders should be aware of limitations:

1. Confidence intervals widen as the forecast horizon increases
2. Unexpected events (weather, trade disruptions) can invalidate predictions
3. Regional variations mean forecasts must be tailored to specific markets
4. Model drift requires regular retraining with new data

9. INTERACTIVE DASHBOARD

9.1 Dashboard Features

To make the forecasting accessible, I developed an interactive Shiny dashboard with:

Feature	Purpose
Forecast Tab	View predicted prices with confidence intervals
Historical Tab	Visualize past trends
Seasonality Tab	Understand monthly patterns
Model Comparison	Compare ARIMA vs Random Forest

9.2 Dashboard Benefits

The dashboard enables stakeholders to:

- Adjust forecast horizon(4-12 weeks) with a simple slider
- Visualize predictions instantly
- Make data-driven decisions in real-time
- Access the forecast without requiring statistical expertise

10. CONCLUSION

10.1 Summary of Findings

This analysis demonstrates that:

1. Avocado prices are highly predictable using time series and machine learning methods
2. Seasonality is the dominant driver, with prices peaking in summer and troughing in spring
3. Regional variation is significant, with major metropolitan areas commanding higher prices
4. ARIMA and Random Forest both perform well, with Random Forest showing slightly better accuracy
5. The alternating weekly pattern identified in the forecast creates clear buy/sell opportunities

10.2 Practical Applications

The insights from this analysis can be applied across the avocado value chain:

- Farmers: Optimize harvest timing based on price forecasts
- Distributors: Plan inventory and shipping schedules
- Retailers: Set pricing strategies and promotion timing
- Investors: Make informed decisions about agricultural commodities

10.3 Future Work

Several extensions could enhance this analysis:

1. Incorporate external factors (weather data, exchange rates, trade policies)
2. Expand to multiple regions with separate models for each
3. Add ensemble methods to combine predictions from multiple models
4. Deploy as a production API for real-time price prediction
5. Include sentiment analysis from social media and news sources

SUPPORTING VISUALIZATIONS

Figure 1: Avocado Price Trend (2015-2018)



Figure 2: Seasonal Pattern by Month

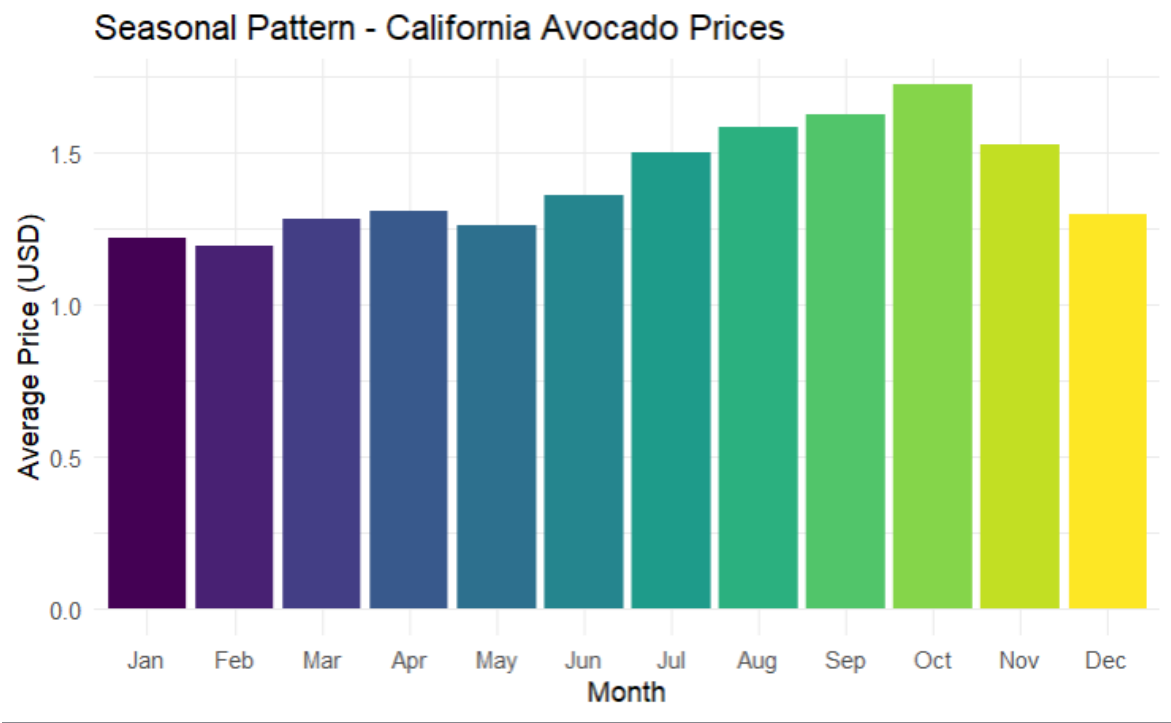


Figure 3: Regional Price Comparison

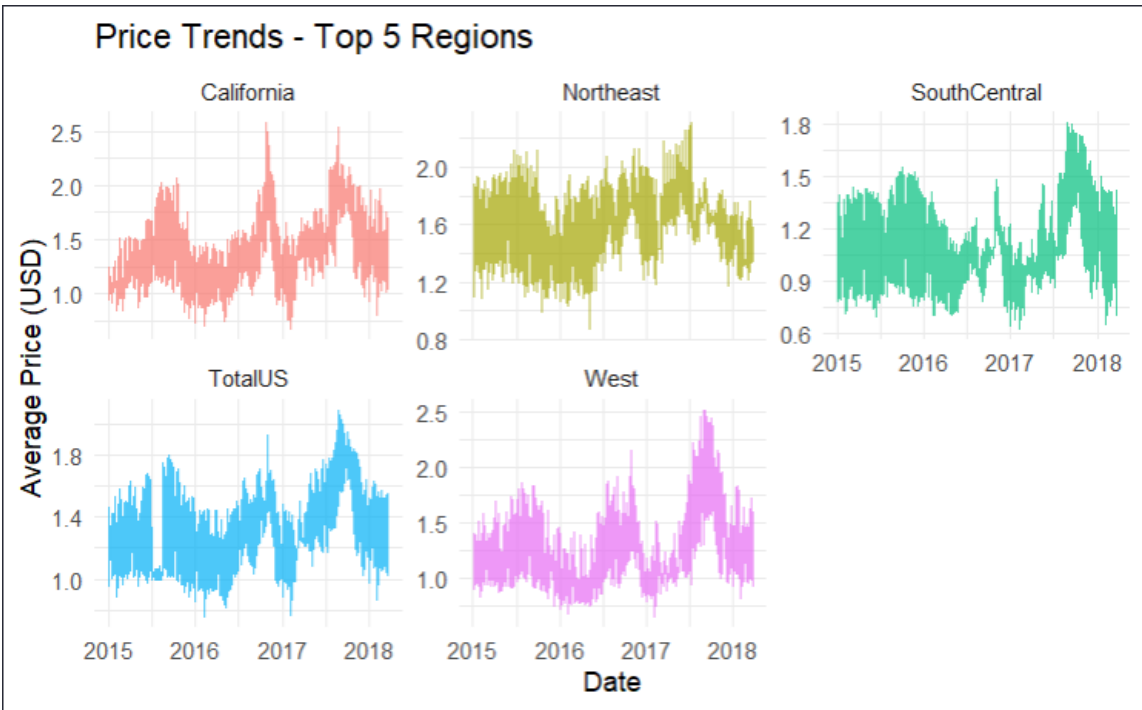


Figure 4: 12-Week Forecast with Confidence Intervals

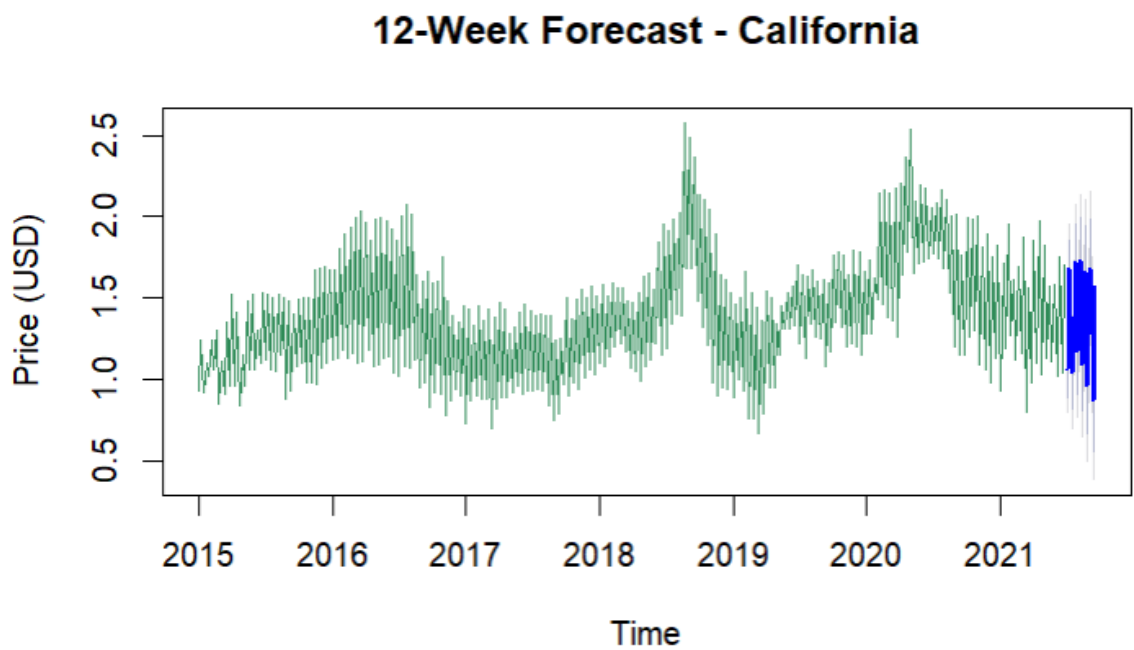
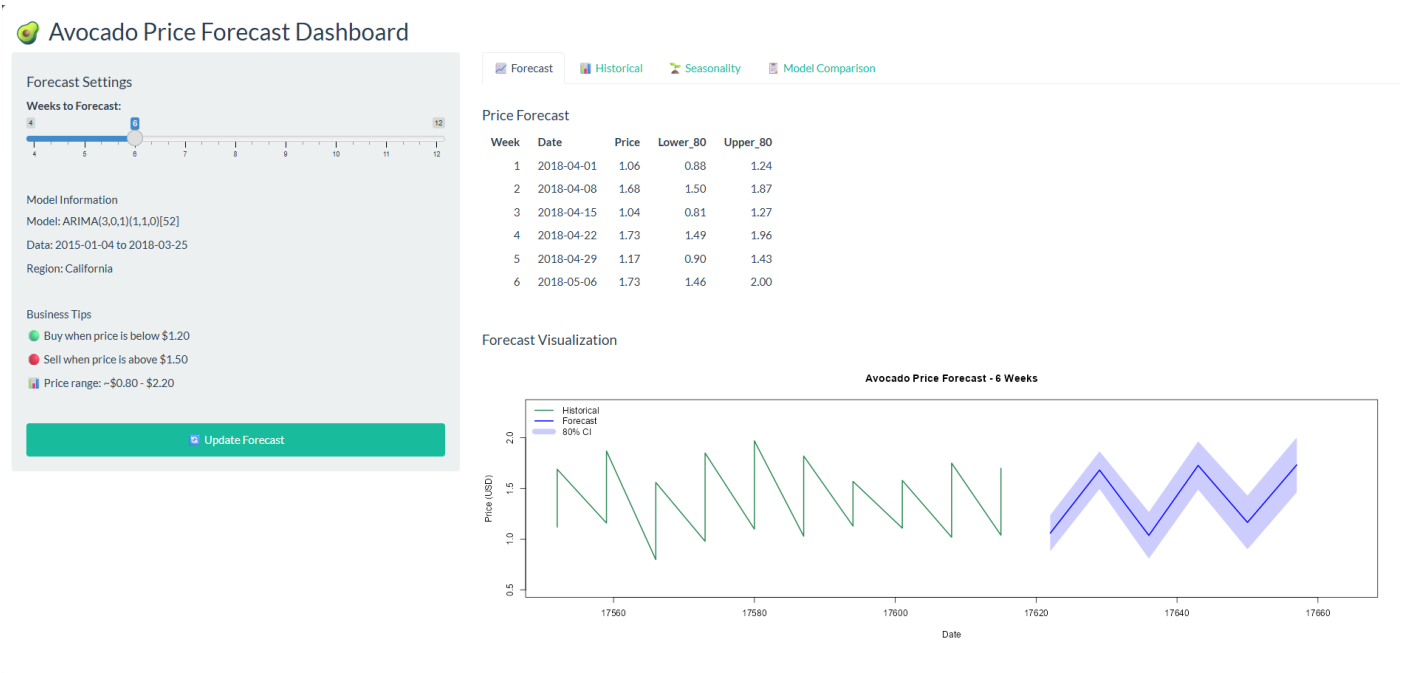


Figure 5: Dashboard Screenshots



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